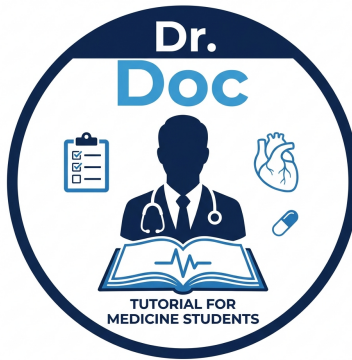




Biochemistry 2014

Introduction to Medicine

Comprehensive Exam Question Bank & Study Guide

All 76 Exam Questions with Highly Detailed, Source-Grounded Explanations

Course Tutorial Series by: **Dr. Doc Tutorial**

Focus Area: Cellular Physiology, Metabolism, & Molecular Genetics

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PROGRAM INTRODUCTION

Welcome to the **Biochemistry 2014: Introduction to Medicine Comprehensive Question Bank**. This study guide compiles all 76 high-yield questions spanning core physiological and biochemical domains including carbohydrate metabolism, lipid and phospholipid structures, amino acid classification, enzyme kinetics, protein secondary structure, eukaryotic and prokaryotic transcription, molecular genetics, and cellular signaling cascades. Every single question is accompanied by its correct choice and an extensive, biochemically rigorous explanation strictly grounded in medical literature, including Marks' Basic Medical Biochemistry: A Clinical Approach and Lippincott's Illustrated Reviews: Biochemistry. This resource has been meticulously designed by **Dr. Doc Tutorial** to ensure maximum retention and a profound conceptual understanding of medical biochemistry for medical students.

How to use this guide: For each question, attempt to solve it independently before reading the detailed explanation. The explanations are structured not only to provide the correct answer but also to explain why alternative choices are incorrect and to reinforce corresponding high-yield clinical correlations. Take notes on key pathways, structures, and clinical vignettes.

"Biochemistry is the chemistry of life, the molecular basis of medical practice. To understand disease, one must first master the delicate molecular pathways that define physiological homeostasis."

Question 1

Which of the following is a carbohydrate?

- A. Arginine
- B. Fructose
- C. Methionine
- D. Maltose

Correct Answer: B. Fructose and D. Maltose

Explanation: Both fructose and maltose are carbohydrates. Fructose is a monosaccharide (a simple ketohexose sugar), while maltose is a disaccharide consisting of two glucose molecules joined by an α -1,4-glycosidic bond. Arginine and methionine are amino acids, not carbohydrates.

Question 2

Glucose consists of

- A. Glycine, arginine, and methionine
- B. Glycine, cysteine, and glutamate
- C. Glycine, histidine, and methionine
- D. Glycine, cysteine, and methionine

Correct Answer: B. Glycine, cysteine, and glutamate (with clarification)

Explanation: Glucose is a monosaccharide sugar ($C_6H_{12}O_6$) and does not consist of amino acids. However, this is a very common exam typo where Glutathione (GSH) was intended. Glutathione is a tripeptide composed of glycine, cysteine, and glutamate. If the question intended to ask for the precursors of creatine, the correct answer would be A. Glycine, arginine, and methionine.

Question 3

The type of homopolysaccharide used in assessing the glomerular filtration rate is?

- A. Agar
- B. Inulin
- C. Pectin
- D. Starch

Correct Answer: B. Inulin

Explanation: Inulin is a plant-derived homopolysaccharide of fructose (fructan). Clinically, inulin is used as the gold standard to measure the glomerular filtration rate (GFR) because it is freely filtered by the glomeruli, and is neither reabsorbed nor secreted by the renal tubules.

Question 4

Which of the following can be produced from Oxidation of galactose?

- A. Mucic acid
- B. Glucuronic acid
- C. Saccharic acid
- D. Gluconic acid

Correct Answer: A. Mucic acid

Explanation: The oxidation of galactose at both C1 (the aldehyde group) and C6 (the terminal primary alcohol group) yields a dicarboxylic acid called mucic acid (also known as galactaric acid). Oxidation of glucose in a similar manner yields saccharic acid.

Question 5

Hyaluronic acid consists of

- A. N-acetyl glucosamine and D-glucuronic acid
- B. N-acetyl galactosamine and D-glucuronic acid
- C. N-acetyl glucosamine and galactose
- D. N-acetyl galactosamine and L-iduronic acid

Correct Answer: A. N-acetyl glucosamine and D-glucuronic acid

Explanation: Hyaluronic acid is a glycosaminoglycan (GAG) composed of repeating disaccharide units of N-acetylglucosamine and D-glucuronic acid. It is a major component of the extracellular matrix, joint fluid, and vitreous humor.

Question 6

Which of the following can be used as the distinguishing test for disaccharides?

- A. Bial's test
- B. Seliwanoff's test
- C. Barfoed's test
- D. Hydrolysis test

Correct Answer: C. Barfoed's test

Explanation: Barfoed's test is used specifically to distinguish monosaccharides from reducing disaccharides. Monosaccharides react much faster with copper ions in the acidic Barfoed's reagent (forming a red precipitate of cuprous oxide within 1 to 2 minutes) compared to reducing disaccharides, which take much longer or require boiling.

Question 7

Which of the following shows the correct combination for Glycosphingolipids?

- A. Ceramide with one or more sugar residues
- B. Glycerol with galactose
- C. Sphingosine with galactose
- D. Sphingosine with phosphoric acid

Correct Answer: A. Ceramide with one or more sugar residues

Explanation: Glycosphingolipids are sphingolipids composed of a hydrophobic ceramide core (sphingosine linked to a fatty acid via an amide bond) joined covalently to one or more sugar residues (such as glucose or galactose) via a glycosidic bond.

Question 8

Genetic information of nuclear DNA is transmitted to the site of protein synthesis by

- A. rRNA
- B. mRNA
- C. tRNA
- D. Polysomes

Correct Answer: B. mRNA

Explanation: Messenger RNA (mRNA) carries the genetic master plan transcribed from nuclear DNA out of the nucleus and to the ribosomes in the cytoplasm, which serve as the direct site of protein synthesis (translation).

Question 9

Identify the different amino acid from the following.

- A. Cysteine
- B. Histidine
- C. Methionine
- D. Threonine

Correct Answer: A. Cysteine

Explanation: In terms of dietary requirements in humans, histidine, methionine, and threonine are classified as essential amino acids. Cysteine is a nonessential amino acid because it can be synthesized endogenously using the carbon skeleton of serine and the sulfur group donated by the essential amino acid methionine.

Question 10

Which of the following statement is true regarding the α -helix of proteins

- A. It is pleated structure
- B. It is Made periodic by disulphide bridges
- C. has secondary periodic structure
- D. Stabilized by hydrogen bonds between NH and CO groups of the main chain

Correct Answer: D. Stabilized by hydrogen bonds between NH and CO groups of the main chain

Explanation: The α -helix is a rigid, spiral secondary structure. It is stabilized by local hydrogen bonds formed between the carbonyl oxygen ($C=O$) of one peptide bond and the amide hydrogen ($N-H$) of another peptide bond located exactly four residues further down the polypeptide backbone.

Question 11

The following antioxidant can be utilized to prevent unpleasant odour and taste in (rancidity)

- A. Lead
- B. Copper
- C. Tocopherol
- D. Ergosterol

Correct Answer: C. Tocopherol

Explanation: Tocopherol (Vitamin E) is a hydrophobic, lipid-soluble antioxidant that acts as a free-radical scavenger in membranes and oils. It prevents the auto-oxidation of unsaturated lipids (lipid peroxidation), which causes the development of unpleasant odors and tastes known as rancidity.

Question 12

The Michaelis constant, K_m is

- A. Numerically equal to $\frac{1}{2} V_{max}$
- B. Dependent on the enzyme concentration
- C. Independent of pH
- D. Numerically equal to the substrate concentration that gives half maximal velocity

Correct Answer: D. Numerically equal to the substrate concentration that gives half

Explanation: The Michaelis constant (K_m) is a key kinetic parameter characteristic of an enzyme-substrate pair. It is numerically defined as the substrate concentration at which the reaction velocity is exactly half of its maximal velocity (V_{max}). It is independent of enzyme concentration.

Question 13

An enzyme promotes a chemical reaction by

- A. Lowering the energy of activation
- B. Causing the release of heat which acts as a primer
- C. Increasing molecular motion
- D. Changing the free energy difference between substrate and product

Correct Answer: A. Lowering the energy of activation

Explanation: Enzymes are highly specific biological catalysts. They accelerate chemical reactions solely by lowering the activation energy barrier (the free energy required to reach the transition state). They do not alter the net free energy change (ΔG) or equilibrium constant of the reaction.

Question 14

In RNA molecule 'Caps'

- A. Allow tRNA to be processed
- B. Are unique to eukaryotic mRNA
- C. Occur at the 3' end of tRNA
- D. Allow tRNA molecule 'Caps'

Correct Answer: B. Are unique to eukaryotic mRNA

Explanation: The 5'-cap (consisting of a 7-methylguanosine residue linked via an unusual 5'-to-5' triphosphate bridge) is a unique modification of eukaryotic mRNA. It does not occur on tRNA or rRNA, nor is it found on prokaryotic transcripts.

Question 15

Except for correct translation of prokaryotic mRNA

- A. Initiator tRNA carries formylated methionine
- B. the tRNA transcript is identical in sequence with mRNA
- C. Except for the replacement of 'T' with 'U', the RNA transcript is identical in sequence with the template strand.
- D. Except for the replacement of 'T' with 'U', the RNA transcript is identical in sequence with coding strand.

Correct Answer: D. Except for the replacement of 'T' with 'U', the RNA transcript is identical

Explanation: RNA transcription uses the template (antisense) strand of DNA to synthesize a complementary RNA transcript. Consequently, the sequence of the synthesized RNA transcript is identical in sequence to the DNA coding (sense/non-template) strand, with the sole exception that uracil (U) replaces thymine (T).

Question 16

Which of the following is not the function of the 5'-capping on mature mRNA?

- A. Protection from degradation
- B. Nuclear exit
- C. Translation termination
- D. Stabilize translation

Correct Answer: C. Translation termination

Explanation: The 5'-capping of mature eukaryotic mRNA serves multiple critical functions: it protects the transcript from 5'-exonuclease degradation, facilitates nuclear export, and stabilizes translation initiation by serving as a ribosome binding site. It has no role in translation termination, which is triggered when a stop codon enters the ribosomal A-site.

Question 17

Which enzyme is used to join nicks in the DNA strand?

- A. Protease
- B. DNA polymerase
- C. DNA ligase
- D. Endonuclease

Correct Answer: C. DNA ligase

Explanation: DNA ligase is the enzyme responsible for joining single-stranded nicks in a double-stranded DNA helix. It catalyzes the formation of a covalent phosphodiester bond between the 3'-hydroxyl group of one nucleotide and the 5'-phosphate group of another during replication, repair, and recombination.

Question 18

DNA replication requires a short strand of _____ primer.

- A. DNA
- B. RNA
- C. Histone
- D. Protein

Correct Answer: B. RNA

Explanation: All replicative DNA polymerases require a pre-existing free 3'-OH group to initiate synthesis. Therefore, DNA replication requires a short complementary RNA primer synthesized de novo by the enzyme primase (or the primase subunit of eukaryotic DNA polymerase α).

Question 19

Which of the following acts as the inducer of the lac operon?

- A. Glucose
- B. Lactose
- C. Galactose
- D. Allolactose

Correct Answer: D. Allolactose

Explanation: The natural physiological inducer of the lac operon is allolactose, an isomer of lactose. It is formed inside the cell by β -galactosidase. Allolactose binds directly to the lac repressor, altering its conformation so that it dissociates from the operator, allowing transcription.

Question 20

Prokaryotic gene transcription is in part controlled by the interaction of repressors and activators. Which of the following does not happen if an activator binds a repressor?

- A. Transcription
- B. Binding of RNA polymerase to the promoter
- C. Binding of the repressor to the promoter
- D. Binding of the allolactose to the repressor

Correct Answer: C. Binding of the repressor to the promoter

Explanation: When an inducer/activator molecule (such as allolactose) binds to the repressor, it causes a conformational change that severely lowers the repressor's affinity for DNA. Thus, the binding of the repressor to the promoter/operator region is prevented (does not happen), allowing RNA polymerase to bind the promoter (B) and initiate transcription (A).

Question 21

Which of the following is correct with regard to DNA replication in eukaryotes?

- A. Single origin and continuous replication
- B. Multiple origins and continuous replication
- C. Single origin, continuous replication and discontinuous replication
- D. Multiple origins, continuous replication and discontinuous replication

Correct Answer: D. Multiple origins, continuous replication and discontinuous replication

Explanation: Eukaryotic chromosomes are large and linear. To complete duplication efficiently, replication initiates at multiple origins of replication along each chromosome. At each replication fork, synthesis is continuous on the leading strand and discontinuous (forming Okazaki fragments) on the lagging strand.

Question 22

If one strand of a double stranded DNA fragment has 20 % cytosine, what is the percentage of adenine?

- A. 20%
- B. 30%
- C. 40%
- D. 50%

Correct Answer: B. 30%

Explanation: Applying Chargaff's Rule to double-stranded DNA: the total $\%C = \%G$ and $\%A = \%T$. Under the standard exam assumption that the double-stranded fragment itself has 20% cytosine (or both strands are symmetric): cytosine (20%) and guanine (20%) make up 40% of the DNA bases. The remaining 60% must be shared equally by adenine and thymine, making adenine 30% of the base composition.

Question 23

Non-coding sequence in mRNA is known as _____

- A. Template
- B. Non-template
- C. Intron
- D. Exon

Correct Answer: C. Intron

Explanation: The non-coding intervening sequences in eukaryotic primary mRNA transcripts (hnRNA) are called introns. They are transcribed but must be spliced out in the nucleus to produce mature coding mRNA consisting only of exons.

Question 24

Genetic information flows from

- A. DNA to DNA
- B. DNA to RNA
- C. RNA to cellular proteins
- D. DNA to cellular proteins

Correct Answer: B. DNA to RNA or D. DNA to cellular proteins

Explanation: The Central Dogma of molecular biology establishes that genetic information flows from DNA to RNA (transcription) and then from RNA to proteins (translation). Thus, DNA to RNA represents the direct informational transcription step, while DNA to cellular proteins represents the full flow of gene expression.

Question 25

Collection of codons

- A. Collection of codons
- B. Collection of amino acids
- C. Collection of purine nucleotides
- D. Collection of pyrimidine nucleotides

Correct Answer: A. Collection of codons

Explanation: The genetic code is defined as the dictionary or collection of three-nucleotide codons that specifies all the amino acids required to synthesize proteins. In this question, due to a clear typographical omission of the term 'Genetic code' in the question stem, option A is the only matching definition.

Question 26

The degeneracy of the genetic code implies that

- A. Some codons do not code for specific amino acid.
- B. Multiple codons can encode the same amino acid.
- C. No anticodons on tRNA molecule.
- D. Specific codon decodes many amino acids

Correct Answer: B. Multiple codons can encode the same amino acid.

Explanation: The genetic code is degenerate (redundant), which means that a single amino acid can be encoded by multiple different triplet codons. For instance, six different codons code for the amino acid leucine.

Question 27

Genetic code is

- A. Overlapping
- B. Non-overlapping
- C. Not universal
- D. Ambiguous

Correct Answer: B. Non-overlapping

Explanation: The genetic code is non-overlapping, meaning that starting from the initiation codon, each nucleotide is read exactly once as part of a single triplet codon. It is also universal and completely unambiguous.

Question 28

In DNA replication, which of the following is required as the first step?

- A. DNA-directed DNA polymerase
- B. Unwinding proteins
- C. DNA polymerase
- D. DNA ligase

Correct Answer: B. Unwinding proteins

Explanation: Before DNA-directed DNA polymerases can synthesize new strands, the double-stranded parental DNA must be separated into single-stranded templates. This crucial first step is performed by unwinding proteins (including helicases, initiator proteins, and single-stranded binding proteins).

Question 29

The genetic code operates through

- A. The protein moiety of DNA
- B. Cistron of DNA
- C. Nucleotide sequence of mRNA
- D. The anticodons of tRNA

Correct Answer: C. Nucleotide sequence of mRNA

Explanation: The genetic code is translated into physical proteins via the ribosome reading the nucleotide sequence of mRNA as a series of three-base codons that specify the precise sequence of amino acids.

Question 30

From DNA the genetic message is transcribed into this molecule

- A. Protein
- B. mRNA
- C. gRNA
- D. rRNA

Correct Answer: B. mRNA

Explanation: In transcription, RNA polymerase read a DNA template strand to synthesize messenger RNA (mRNA), which carries the transcribed genetic message to the cytoplasm for translation.

Question 31

Which one of the following compounds has a double helical structure?

- A. Ribonucleic acid
- B. Deoxyribonucleic acid
- C. Flavine-adenine dinucleotide
- D. Nicotinamide adenine dinucleotide

Correct Answer: B. Deoxyribonucleic acid

Explanation: Deoxyribonucleic acid (DNA) possesses a highly stable double helical structure consisting of two complementary, antiparallel polynucleotide strands wound around a central axis.

Question 32

The structural stability of the double helix of DNA is attributed largely to

- A. Hydrogen bonding between adjacent purine bases.
- B. Hydrophobic bonding between stacked purine and pyrimidine bases.
- C. Hydrogen bonding between adjacent pyrimidine bases.
- D. Hydrogen bonding between purine and pyrimidine bases.

Correct Answer: D. Hydrogen bonding between purine and pyrimidine bases.

Explanation: The complementary double strands of DNA are held together and structurally stabilized primarily by hydrogen bonding between specific complementary purine and pyrimidine bases (A pairs with T via 2 hydrogen bonds; G pairs with C via 3 hydrogen bonds).

Question 33

DNA is

- A. Usually present in tissues as a nucleoprotein and cannot be separated from its protein component.
- B. A long chain polymer in which the inter-nucleotide linkages are of the diester type between C-3' and C-5'.
- C. Different from RNA since in the latter the inter-nucleotide linkages are between C-2' and C-5'
- D. Hydrolysable by weak alkali (pH 9 to 100°C).

Correct Answer: B. A long chain polymer in which the inter-nucleotide linkages are of the

Explanation: DNA is a linear, unbranched polymer of nucleotides where the individual monomer units are covalently joined by 3' $\rightarrow$ 5' phosphodiester linkages. Unlike RNA, DNA is highly resistant to alkali-catalyzed hydrolysis.

Question 34

An mRNA has the sequence of AAUUGGCU. The sequence of the non-template strand of the DNA that gave rise to this sequence is which one of the following?

- A. AGCCAATT
- B. AAUUGGCU
- C. AATTGGCT
- D. TTAACCGA
- E. UUAACCGA

Correct Answer: C. AATTGGCT

Explanation: The non-template (coding/sense) strand of DNA is identical in sequence and polarity to the synthesized mRNA transcript, except that DNA contains thymine (T) instead of uracil (U). Thus, for mRNA `5'-AAUUGGCU-3'`, the coding DNA sequence is `5'-AATTGGCT-3'`.

Question 35

The function of a repressor protein in an operon system is to prevent synthesis by binding to

- A. The ribosome
- B. A specific region of the operon preventing transcription of structural genes
- C. The RNA polymerase
- D. A specific region of the mRNA preventing translation to protein

Correct Answer: B. A specific region of the operon preventing transcription of structural

Explanation: A repressor protein regulates gene transcription in bacterial operons by binding to a specific DNA sequence called the operator. This binding physically blocks RNA polymerase from transcribing the downstream structural genes.

Question 36

Lac operon of *E. coli* contains _____ in continuity.

- A. Regulator and operator genes only
- B. Operator and structural genes only
- C. Regular and structural genes only
- D. Regulator, operator and structural genes

Correct Answer: D. Regulator, operator and structural genes

Explanation: The lac operon is a functional genetic unit composed of regulatory element genes (the $\text{\$lacI\$}$ regulator with its promoter), control regions (the operator and promoter), and the three structural genes ($\text{\$lacZ\$}$, $\text{\$lacY\$}$, $\text{\$lacA\$}$) transcribed together in continuity.

Question 37

An mRNA of eukaryotes can code for

- A. Only one polypeptide
- B. Two different polypeptides
- C. Three different polypeptides
- D. Many different polypeptides, as long as it is not degraded.

Correct Answer: A. Only one polypeptide

Explanation: Eukaryotic messenger RNA is monocistronic, meaning it carries the coding sequence to produce only one polypeptide chain from a single translation start site. Prokaryotic mRNA, by contrast, is often polycistronic.

Question 38

In DNA, genetic information is located in

- A. Purine bases
- B. Pyrimidine bases
- C. Purine and pyrimidine bases
- D. Sugar backbone

Correct Answer: C. Purine and pyrimidine bases

Explanation: The genetic information of an organism is stored precisely within the linear sequence of nitrogenous bases. These bases belong to two chemical families: purines (adenine and guanine) and pyrimidines (cytosine and thymine).

Question 39

The polyadenylation (PolyA) tail on the 3' end of mRNA is thought to function as:

- A. A transcriptional signal for RNA polymerase
- B. A degradation signal for RNase digestion
- C. A stabilization sequence during RNA translation
- D. A precursor sequence which is selectively removed after mRNA capping

Correct Answer: C. A stabilization sequence during RNA translation

Explanation: The poly(A) tail on eukaryotic mRNA serves to protect the transcript from rapid enzymatic degradation in the cytosol and functions as a binding site for stabilizing proteins that interact with the ribosome to facilitate translation initiation.

Question 40

A patient with manic depressive disorder is treated with lithium, which slows the turnover of inositol phosphates and the phosphatidyl inositol derivatives in cells. Which of the following protein kinases is most directly affected by this drug?

- A. Protein Kinase M
- B. Protein Kinase G
- C. Protein Kinase A
- D. Protein Kinase C

Correct Answer: D. Protein Kinase C

Explanation: Lithium blocks the recycling of inositol, leading to depletion of membrane phosphatidylinositol 4,5-bisphosphate (PIP_2). Normally, PIP_2 is cleaved to generate IP_3 and diacylglycerol (DAG). Because DAG is the primary activator of Protein Kinase C (PKC), lithium treatment most directly dampens PKC signaling.

Question 41

In Eukaryotic their DNA in the nucleus is coiled in to a dense mass called

- A. Histone protein
- B. Nucleolus
- C. Nucleoplasm
- D. None

Correct Answer: D. None (The correct answer is Chromatin)

Explanation: In eukaryotic nuclei, genomic DNA associates with basic histone proteins to coil into a highly organized, dense nucleoprotein complex called chromatin. Since 'chromatin' is not listed among options A, B, or C, the correct option is D. None.

Question 42

Among the following Enzyme except one all of them found in inner membrane of Mitochondria

- A. Ubiquinone
- B. All translocases
- C. All dehydrogenases
- D. Cytochromes b

Correct Answer: C. All dehydrogenases

Explanation: The inner mitochondrial membrane houses respiratory complexes containing cytochromes, ubiquinone, and various translocases. However, most mitochondrial dehydrogenases (like those of the TCA cycle or fatty acid oxidation) are soluble enzymes located inside the mitochondrial matrix (with succinate dehydrogenase as the notable inner membrane exception).

Question 43

Which of the following statements is true?

- A. When signal molecules first bind to receptor tyrosine kinases, the receptors phosphorylate a number of nearby molecules
- B. In response to some G-protein-mediated signals, a special type of lipid molecule associated with the plasma membrane is cleaved to form IP3 and calcium.
- C. Protein kinase A activation is one possible result of signal molecules binding protein-linked receptors.
- D. All

Correct Answer: C. Protein kinase A activation is one possible result of signal molecules

Explanation: The binding of extracellular signaling ligands to G-protein coupled (heptahelical) receptors can activate adenylyl cyclase, producing cAMP which directly activates Protein Kinase A (PKA). Statement B is false because PIP_2 is cleaved into IP_3 and diacylglycerol (DAG)—calcium is not directly cleaved from the membrane. Statement A is false because RTKs autophosphorylate themselves initially.

Question 44

From the perspective of the cell receiving the message, the three stages of cell signaling are

- A. DNA, paracrine, local, and synaptic response
- B. signal reception, signal translocalization, and cellular responses
- C. signal reception, nucleotide disintegration, and new cell generation
- D. signal reception, signal transduction, and cellular response

Correct Answer: D. signal reception, signal transduction, and cellular response

Explanation: The cellular perception and processing of extracellular signals proceed in a strictly coordinated sequence of three stages: reception (ligand binding to receptor), transduction (propagation of intracellular messengers), and cellular response (the final physiological change).

Question 45

A G-protein receptor with GTP bound to it ____

- A. in its active state
- B. signals a protein to maintain its shape and conformation
- C. directly affects gene expression
- D. in its inactive state

Correct Answer: A. in its active state

Explanation: G-proteins act as intracellular molecular switches. In their inactive state, they bind GDP. Upon ligand binding to the G-protein coupled receptor, they exchange GDP for GTP, transitioning into their fully active state to regulate downstream effectors.

Question 46

Which of the following cell organelles is called a suicidal bag?

- A. Nucleus
- B. Golgi bodies
- C. Lysosomes
- D. Mitochondria
- E. All except 'D'

Correct Answer: C. Lysosomes

Explanation: Lysosomes are spherical, single-membrane bound organelles containing highly acidic hydrolytic enzymes. If a cell undergoes programmed autolysis or is severely damaged, lysosomes rupture and release these acidic hydrolases globally into the cytoplasm, digesting the cell from within.

Question 47

Which of the following nucleotide is used as secondary messenger

- A. cAMP
- B. NADPH
- C. ATP
- D. GMP

Correct Answer: A. cAMP

Explanation: Cyclic adenosine monophosphate (cAMP) is a cyclic nucleotide derived from ATP by the action of the membrane-bound enzyme adenylyl cyclase. It is one of the most widely used intracellular second messengers in biology.

Question 48

Diseases caused by defects in DNA repair systems put the patient at risk for developing

- A. Cancers. DNA repair mechanisms can be best described by which one of the following?
- B. Proofreading works as the bases are paired and eliminates all base pairing errors.
- C. After replication, no further repairs are possible.
- D. Genes that produce mRNA have a unique repair system.
- E. DNA polymerases (both repair and replicative) can use any 3'OH for elongation.

Correct Answer: D. Genes that produce mRNA have a unique repair system.

Explanation: Eukaryotes have a specialized, transcription-coupled DNA repair system (TCR) that specifically targets and prioritizes the template strands of actively transcribing genes (which produce mRNA) to ensure crucial proteins are transcribed without error. Statement B is incorrect because proofreading does not catch 100% of replication errors.

Question 49

DNA replication is a different process than DNA repair. Mutations in DNA repair enzymes can lead to disease—especially certain forms of cancer. Primase is not required during DNA repair processes because of which one of the following?

- A. All of the primase is associated with replication origins.
- B. RNA would be highly mutagenic at a repair site.
- C. Repair DNA polymerases do not require a primer.
- D. Replicative DNA polymerases do not require a primer.
- E. DNA Polymerases (Both repair and replicative) can use any 3-OH for elongation

Correct Answer: C. Repair DNA polymerases do not require a primer.

Explanation: Unlike replicative DNA polymerases which must start synthesis de novo, repair DNA polymerases do not require primase to synthesize an RNA primer. This is because the excision of damaged DNA leaves a single-stranded gap adjacent to an already existing, intact double-stranded DNA segment, which provides a free 3'-OH end that the repair polymerase uses directly to begin elongation.

Question 50

Nucleotide CTP is used for

- A. Principal cellular energy source
- B. Metabolic reactions of lipids
- C. Nucleotide sequence of mRNA
- D. The anticodons of tRNA

Correct Answer: B. Metabolic reactions of lipids

Explanation: While ATP serves as the primary cellular energy currency, cytidine triphosphate (CTP) acts as a high-energy activation intermediate specifically in the biosynthesis of phospholipids and complex lipids (such as the formation of CDP-choline and CDP-diacylglycerol).

Question 51

All of the following may be used as expression vectors except

- A. Plasmid
- B. Bacteriophage
- C. Cosmids
- D. E. coli

Correct Answer: D. E. coli

Explanation: An expression vector is a DNA molecule (such as a plasmid, bacteriophage, or cosmid) engineered to join with and carry a foreign DNA segment into a host cell. Escherichia coli (E. coli) is a prokaryotic host cell used to replicate and express vectors, not a vector itself.

Question 52

The enzyme responsible for the synthesis of RNA primer in eukaryotes

- A. DNA polymerase α
- B. DNA polymerase β
- C. DNA polymerase γ
- D. Topoisomerases

Correct Answer: A. DNA polymerase α

Explanation: In eukaryotic cells, DNA polymerase α (Pol α) is a multi-subunit enzyme that possesses intrinsic primase activity, making it responsible for the synthesis of the short RNA primer that initiates DNA replication.

Question 53

Restriction Endonucleases

- A. Cut RNA chains at specific locations
- B. Excise introns from hnRNA
- C. Remove Okazaki fragments
- D. Act as defensive enzymes to protect the host bacterial DNA from DNA of foreign organisms

Correct Answer: D. Act as defensive enzymes to protect the host bacterial DNA from DNA

Explanation: In nature, restriction endonucleases are synthesized by bacterial cells to act as a defensive mechanism that 'restricts' the growth of invading foreign DNA, such as bacteriophages, by cleaving it at specific palindromic sequences.

Question 54

Which one of the following statement is incorrect about codon and anti-codon?

- A. Both codon and anti-codon consist of sequence of three nucleotides.
- B. Codons are present on DNA and anti-codon is present on tRNA.
- C. Codon and anticodon are complementary in nature
- D. None of the above

Correct Answer: B. Codons are present on DNA and anti-codon is present on tRNA.

Explanation: Codons are defined as three-nucleotide sequences present on messenger RNA (mRNA). While the coding strand of DNA mirrors the sequence of mRNA (with T instead of U), codons are represented on the transcribed mRNA template, whereas the complementary anticodon is located on transfer RNA (tRNA).

Question 55

Certain drugs can intercalate between DNA bases and alter the backbone of the DNA. The backbone of a DNA strand is composed of which of the following? Choose the one best answer.

- A. Phosphates and sugars
- B. Bases and phosphates
- C. Nucleotides and sugars
- D. Phosphates and nucleosides
- E. Sugars and bases

Correct Answer: A. Phosphates and sugars

Explanation: The structural backbone of a DNA strand is composed of repeating deoxyribose sugar and phosphate groups linked together covalently by $3' \rightarrow 5'$ phosphodiester bonds. The nitrogenous bases project inward from this backbone.

Question 56

Some chemotherapeutic drugs alter the ability of DNA polymerase to faithfully replicate. For the DNA sequence 5'-ATCGATCGATCGATCG-3', which one of the following represents the sequence and polarity of the complementary strand?

- A. 5'-ATCTATCGATCGATCG-3'
- B. 3'-ATCGATCGATCGATCG-5'
- C. 5'-CGAUCGAUCGAUCGAU-3'
- D. 5'-CGATCGATCGATCGAT-3'
- E. 3'-CGATCGATCGATCGAT-5'

Correct Answer: D. 5'-CGATCGATCGATCGAT-3'

Explanation: DNA strands are complementary and antiparallel. Using base-pairing rules (A-T and G-C), the complementary strand for '5'-ATCGATCGATCGATCG-3'' in the opposite polarity is '3'-TAGCTAGCTAGCTAGC-5''. Written in the standard '5' $\rightarrow$ '3' direction, this is '5'-CGATCGATCGATCGAT-3''.

Question 57

One of the main sources of nonvolatile acid in the body is sulfuric acid generated from sulfur-containing compounds in ingested food or from the metabolism of the sulfur-containing amino acids. Which of the following amino acids would lead to sulfuric acid formation?

- A. Cysteine and isoleucine
- B. Cystine and alanine
- C. Cysteine and methionine
- D. Methionine and isoleucine
- E. Isoleucine and alanine

Correct Answer: C. Cysteine and methionine

Explanation: Cysteine (along with its dimer cystine) and methionine are the sulfur-containing amino acids. The metabolic breakdown of these amino acids in the human body generates sulfuric acid, a major source of nonvolatile metabolic acid.

Question 58

Phosphorylation of proteins is an important component of signal transduction. Protein kinases phosphorylate proteins only at certain hydroxyl groups on amino acid side chains. Which of the following groups of amino acids contain side-chain hydroxyl groups and could be a potential substrate for a protein kinase?

- A. Alanine, glutamine, and lysine
- B. Serine, threonine, and tyrosine
- C. Threonine, phenylalanine, and arginine
- D. Cysteine, tryptophan, and glycine
- E. Alanine, asparagine, and serine

Correct Answer: B. Serine, threonine, and tyrosine

Explanation: Protein kinases selectively catalyze the transfer of phosphate groups from ATP to the free, polar hydroxyl (-OH) groups on amino acid side chains. The only three standard amino acids containing side-chain hydroxyl groups are serine, threonine, and tyrosine.

Question 59

Proteins, which are composed of amino acids, help transport lipids in the bloodstream. These proteins need to be able to cluster with other nonpolar molecules and exclude water. Which of the following would best describe the side chains of these amino acids in the lipid transport proteins?

- A. A more positive hydrophilic index
- B. A more negative hydrophilic index
- C. A more hydrophobic index
- D. A pKa of the alpha-carboxylic acid group of approximately 2
- E. A pKa of the alpha-amino group of approximately 9.5

Correct Answer: C. A more hydrophobic index

Explanation: The hydrophobic index measures the hydrophobicity of amino acid side chains. A more hydrophobic index (or a more positive hydrophobic index) describes side chains that cluster with nonpolar lipids and exclude water. While backbone carboxylic and amino groups have pKa values of ~2 and ~9.5, they do not contribute to the hydrophobic effect.

Question 60

A patient was born with a congenital mutation in an enzyme that severely affected its ability to bind an activation-transfer coenzyme. As a consequence, which one of the following is most likely to occur?

- A. The enzyme will be unable to bind the substrate of the reaction.
- B. The enzyme will be unable to form the transition-state complex.
- C. The enzyme will normally use a different activation transfer coenzyme.
- D. The enzyme will normally substitute the functional group of an active-site amino acid residue for the coenzyme.
- E. The reaction may be carried out by the free coenzyme, provided the diet carries an adequate amount of its vitamin precursor.

Correct Answer: B. The enzyme will be unable to form the transition-state complex.

Explanation: Activation-transfer coenzymes participate directly in catalysis by forming a covalent transition-state complex with the substrate. A mutant enzyme unable to bind its coenzyme can still bind its substrate but cannot form the transition-state complex to catalyze the reaction.

Question 61

In order for lipid-based hormones such as testosterone or estrogen to be transported in the bloodstream, they are bound to and transported by water-soluble proteins. In order to be water-soluble, the transporting protein contains which one of the following amino acids on its surface in contact with aqueous blood?

- A. Valine
- B. Alanine
- C. Leucine
- D. Phenylalanine
- E. Glutamine

Correct Answer: E. Glutamine

Explanation: To maintain water solubility, the outer surfaces of transport proteins must expose polar, hydrophilic amino acids to interact with the aqueous blood. Glutamine is a polar, hydrophilic amino acid. In contrast, valine, alanine, leucine, and phenylalanine are nonpolar and hydrophobic.

Question 62

Methanol (CH_3OH) is converted by ADH to formaldehyde (CH_2O), a compound that is highly toxic to humans. Patients who have ingested toxic levels of methanol are sometimes treated with ethanol ($\text{C}_2\text{H}_5\text{OH}$) to inhibit methanol oxidation by ADH. Which one of the following is a competitive analog of methanol and might therefore be an effective treatment?

- A. Ethanol is a structural analog of methanol that can be expected to compete with methanol for binding to the ADH enzyme
- B. In vivo studies indicated that the K_m for the ADH for the oxidation of methanol is competitively reduced by ethanol
- C. Ethanol has no effective inhibitory effect of methanol oxidation regardless of the concentration of methanol
- D. Ethanol can be expected to inhibit the enzyme by binding to the formaldehyde-binding site on the enzyme, even though it cannot bind at the substrate binding site for methanol

Correct Answer: A. Ethanol is a structural analog of methanol that can be expected to

Explanation: Ethanol serves as a structural analog of methanol and acts as a competitive inhibitor of alcohol dehydrogenase (ADH). By competing for binding at the ADH active site, ethanol slows the rate of methanol oxidation to highly toxic formaldehyde, allowing methanol to be safely excreted.

Question 63

In tRNA molecules, there is a 'loop' which contains a minor base:

- A. Cytosine
- B. Dihydrocytosine
- C. Thymine
- D. Dihydrouracil
- E. Uracil

Correct Answer: D. Dihydrouracil

Explanation: tRNA molecules contain a high percentage of unusual, post-transcriptionally modified bases. The D-loop is named for the presence of dihydrouracil (dihydrouridine), which is formed by the enzymatic reduction of uracil.

Question 64

The anticodon region is an important part of the structure of:

- A. r-RNA
- B. m-RNA
- C. hn-RNA
- D. t-RNA
- E. Z-DNA

Correct Answer: D. t-RNA

Explanation: The anticodon region is a specific trinucleotide loop sequence located in transfer RNA (tRNA). It base-pairs with the complementary codon on mRNA during translation to ensure correct amino acid insertion.

Question 65

Which of these statements is not true regarding the cell membrane?

- A. The phosphate heads are polar
- B. It is a phospholipid bilayer
- C. The hydrophilic tails face outwards
- D. It contains integral proteins

Correct Answer: C. The hydrophilic tails face outwards

Explanation: The cell membrane is a phospholipid bilayer containing integral and peripheral proteins. Phospholipids are amphipathic: the polar, hydrophilic phosphate heads face outwards toward the aqueous environments, while the hydrophobic hydrocarbon tails face inwards, forming the core. Therefore, statement C is false (the tails are hydrophobic, not hydrophilic, and they face inward).

Question 66

An antibiotic is developed that is a close structural analog of a substrate of an enzyme that participates in cell wall synthesis in bacteria. This binding of the antibiotic reduces overall enzyme activity, but such activity can be restored if more substrate is added. The binding of the antibiotic to the enzyme is not via a covalent bond, nor does the enzyme alter the structure of the antibiotic. Which one of the following would best describe this antibiotic?

- A. It is a suicide inhibitor.
- B. It is an irreversible inhibitor.
- C. It is a competitive inhibitor.
- D. It is an uncompetitive inhibitor.
- E. p53 directly phosphorylates the transcription factor jun

Correct Answer: C. It is a competitive inhibitor.

Explanation: The antibiotic acts as a competitive inhibitor because it is a structural analog of the substrate, binds reversibly (non-covalently), and its inhibitory effect can be completely overcome by adding more substrate.

Question 67

Mutations in the p53 gene are the most common aberrations in spontaneous human cancers. The normal p53 protein affects the cell cycle by

- A. Inducing cell cycle arrest and apoptosis in response to DNA damage
- B. Inducing the phosphorylation of Rb in response to mitogens
- C. Directly inhibiting Cdk inhibitors in response to cell-cell contact and other growth-inhibiting stimuli
- D. Increasing the activity of cyclin-dependent protein kinases by inducing their phosphorylation
- E. Binding and thereby inactivating the products of many proapoptotic genes

Correct Answer: A. Inducing cell cycle arrest and apoptosis in response to DNA damage

Explanation: The p53 tumor suppressor protein ('guardian of the genome') accumulates in response to DNA damage. It acts as a transcription factor, upregulating the Cdk inhibitor p21 to arrest the cell cycle, or inducing pro-apoptotic genes (like Bax) if damage is irreparable.

Question 68

Which one of the following is not true about M-check-point

- A. Anaphase promoting complex/ Cyclosome is key regulator
- B. Triggered mainly by activation of specific cyclin-Cdk complexes via phosphorylation or dephosphorylation
- C. Progression of metaphase-to-anaphase transition triggered by cyclical proteolysis
- D. Checks whether alignment of every chromosomes on spindle
- E. All are true

Correct Answer: B. Triggered mainly by activation of specific cyclin-Cdk complexes via

Explanation: The spindle assembly checkpoint (M-checkpoint) regulates the metaphase-to-anaphase transition. It is controlled by the Anaphase-Promoting Complex/Cyclosome (APC/C) via targeted proteolysis of cyclins and securin. It is not triggered by activating Cdks; rather, it results in Cdk inactivation.

Question 69

The protein that holds the two sister chromatids in place until they are separated is ____.

- A. Kinetochore
- B. Centromere
- C. Cohesin
- D. Securin

Correct Answer: C. Cohesin

Explanation: Cohesin is a multi-protein complex that forms a ring around sister chromatids, holding them together from the S-phase of replication until their separation during anaphase by the enzyme separase.

Question 70

What is the name of the protein signaling molecule that alters glucose uptake, and where would its receptors be located?

- A. Insulin; many different cell types that use glucose for fuel
- B. PDGF; the blood
- C. NGF; the nerves involved in simple reflexes
- D. Insulin; beta cells of the pancreas

Correct Answer: A. Insulin; many different cell types that use glucose for fuel

Explanation: Insulin regulates glucose uptake by promoting the recruitment of GLUT-4 glucose transporters to the plasma membranes of target tissues, primarily skeletal muscle and adipose tissue. Receptors are located on these cell types.

Question 71

The cell's structural unit that serves as microtubules organizing center and facilitates cell division during mitosis is _____.

- A. Centriole
- B. Ribosome
- C. Endoplasmic Reticulum
- D. All

Correct Answer: A. Centriole

Explanation: The centrioles (located within the centrosome) are the primary microtubule organizing centers (MTOCs) in animal cells. They organize the spindle apparatus required to align and separate chromosomes during mitosis.

Question 72

Which of the following processes involve 'the combining of a message from one signaling molecule with that of another to either enhance or inhibit a cellular effect'?

- A. Signal transduction
- B. Signal integration
- C. Signal amplification
- D. Signal reception

Correct Answer: B. Signal integration

Explanation: Signal integration is the process by which a cell receives multiple distinct signals from different pathways and integrates them to produce a single, coordinated cellular response.

Question 73

Sex chromosomal Monosomy

- A. Edward syndrome
- B. Patau syndrome
- C. Down syndrome
- D. Turner syndrome
- E. Cri-du-chat syndrome

Correct Answer: D. Turner syndrome

Explanation: Turner syndrome is a sex chromosomal monosomy (45,X0) in females. Down, Patau, and Edward syndromes are autosomal trisomies (trisomy 21, 13, and 18 respectively).

Question 74

In terms of cell communication, what do bacterial pathogens such as cholera and anthrax have in common?

- A. They destroy the receptors for key signaling molecules
- B. They block the normal functioning of signal transduction mechanisms
- C. They alter the chemical structure of key signaling molecules
- D. They prevent the production of key signaling molecules

Correct Answer: B. They block the normal functioning of signal transduction mechanisms

Explanation: Bacterial pathogens like *Vibrio cholerae* and *Bacillus anthracis* produce toxins that chemically modify crucial signal transduction proteins (e.g., ADP-ribosylation of G α s by cholera toxin keeps adenylyl cyclase permanently active), blocking normal signal regulation.

Question 75

Each of the following numbered processes are involved in signal transduction pathways: I. Response II. Amplification III. Transduction IV. Reception. Which of the following represents the sequence of events in a typical signal transduction pathway?

- A. IV, II, III, I
- B. IV, I, III, II
- C. I, II, III, IV
- D. IV, III, II, I

Correct Answer: D. IV, III, II, I

Explanation: A signal transduction pathway proceeds in the following sequence: Reception of the signal (IV), Transduction of the signal across the membrane (III), Amplification of the signal cascade (II), and final cellular Response (I).

Question 76

The complete expression of the lac operon requires _____

- A. Lactose
- B. Allolactose
- C. Lactose and cAMP
- D. Allolactose and cAMP

Correct Answer: D. Allolactose and cAMP

Explanation: Complete transcription of the lac operon requires: (1) Allolactose (the inducer) to bind and inactivate the repressor, freeing the operator, and (2) cyclic AMP (cAMP), which increases in the absence of glucose and binds the Catabolite Activator Protein (CAP/CRP) to recruit RNA polymerase to the promoter. SOURCE REFERENCES 1. Lieberman, M., & Peet, A. (2018). Marks' Basic Medical Biochemistry: A Clinical Approach, 5th Edition. Wolters Kluwer. 2. Harvey, R. A., & Ferrier, D. R. (2011). Lippincott's Illustrated Reviews: Biochemistry, 5th Edition. Wolters Kluwer / Lippincott Williams & Wilkins. 3. Dr. Doc Tutorial Series. (2014). Lecture Notes & Handouts: Introduction to Medicine - Biochemistry Module. Dr. Doc Tutorial.

